

Custom ASIC and System Architectures for 1T-10T LLM Inference

Executive summary

Inference for 1T-10T parameter LLMs is dominated by *data movement*: weights, activations, and especially KV-cache traffic at long context and high concurrency. Vendor architectures differentiate primarily by where weights live during decode (HBM vs on-chip SRAM vs near-memory), how predictably they schedule token pipelines, and how efficiently they scale collective communication for tensor/expert parallelism. ¹

Key takeaways from the most relevant primary sources:

- **Cerebras** pushes the extreme end of “*weights on chip*” via wafer-scale SRAM and fabric bandwidth. Its WSE-3-based systems advertise **44 GB on-chip SRAM**, **125 PFLOPS**, and **PB/s-scale on-chip bandwidth**, and they publish application-facing inference benchmarks such as **2,100 tokens/s on Llama 3.1 70B** and **969 tokens/s on Llama 3.1 405B** (including **~240 ms TTFT** and **128K context** in a reported record run). ²
- **NVIDIA** remains the most complete hardware-software “AI factory” stack, with multi-GPU NVLink domains (e.g., NVL72) and a rapidly evolving inference software layer (TensorRT-LLM + Dynamo) that adds features like paged/quantized KV cache, disaggregated prefill/decode, and KV-aware routing. Public NVIDIA materials explicitly position Blackwell systems for **real-time trillion-parameter inference**, cite **~130 TB/s** of GPU communications inside a 72-GPU NVLink domain, and describe architectural support for **FP4 inference** and large-scale NVLink scaling (up to hundreds of GPUs). ³
- **Groq** focuses on deterministic, compiler-scheduled pipelines (“programmable assembly line”), combining **hundreds of MB of SRAM per chip** and claimed **~80 TB/s** on-die bandwidth with a direct chip-to-chip protocol and static scheduling. Groq also explicitly frames its approach as enabling fast tensor-parallel decode for very large models (it cites “Kimi K2” as a 1T-parameter example in a technical blog). ⁴
- **d-Matrix** targets the decode bottleneck with a **digital in-memory compute (DIMC)** chiplet platform (Corsair) and publishes both architectural specs (e.g., chiplets, SRAM/LPDDR, TOPS/W, compression) and *token/s projections* (e.g., a table projecting **~60k tokens/s** for Llama3 8B and **~30k tokens/s** for Llama3 70B at specified batch/sequence settings). ⁵
- **Etched** (Sohu) represents the “transformer-only ASIC” extreme. Publicly available details are mostly claims and limited disclosures (including from interviews/press coverage and official social posts), which creates a verification gap versus the above vendors’ more concrete published specs and system documentation. ⁶

Across all vendors, the critical open problem for 1T-10T inference is **how to sustain low latency at long context while scaling to many devices without turning the system into a networking/coordination machine**. The most promising direction is *hybrid disaggregation* (separating prefill and decode) plus *memory-tiered* KV-cache management, paired with topology-aware scheduling and workload-aware routing.

This is now explicitly supported in modern GPU software stacks (NVIDIA Dynamo + TensorRT-LLM) and appears in multiple vendors' system narratives. ⁷

What changes at 1T-10T parameter inference

A 1T-parameter dense model's weight memory is enormous: as a rule of thumb, weight footprint scales with $\text{\#params} \times \text{bytes/param}$ (for example, NVIDIA's own inference guidance illustrates that a 7B FP16 model is ~14 GB, implying linear scaling with parameters and precision). ⁸

- **1T params:** ~2 TB at FP16/BF16; ~1 TB at 8-bit; ~0.5 TB at 4-bit (ignoring metadata/packing). (Derived from the linear scaling rule.) ⁸
- **10T params:** ~20 TB at FP16/BF16; still multi-terabyte even at 4-bit. (Same scaling.) ⁸

Two inference phases stress hardware differently:

- **Prefill** (prompt processing) tends to be more compute-efficient (higher arithmetic intensity) and can benefit from large matmuls and batching; multiple published system efforts now treat it as *separable* from decode. ⁹
- **Decode** (token generation) repeatedly touches weights and KV cache, and becomes memory- and communication-sensitive as contexts grow. d-Matrix explicitly calls token generation "memory bound," highlighting KV-cache growth and small-batch SLO constraints; Cerebras similarly frames decode as bandwidth-intensive because weights must be fetched for each generated token in conventional designs. ¹⁰

KV cache is the other "silent giant." NVIDIA summarizes KV-cache memory as scaling with layers, heads, head dimension, precision, and tokens; batching multiplies the KV footprint because each request's KV must be stored separately. ⁸ Long context therefore couples directly to memory capacity *and* memory bandwidth, particularly under concurrency. d-Matrix's Hot Chips material explicitly links longer KV cache to reduced arithmetic intensity and stricter memory limits. ¹¹

From these constraints, 1T-10T inference architectures typically rely on some combination of:

- **Aggressive quantization formats** (FP8/FP4, MX formats, INT8/INT4 variants) to reduce weight and KV footprint; NVIDIA frames Blackwell's Transformer Engine as enabling FP4 inference, and d-Matrix promotes MXINT8/MXINT4 and block floating point tied to OCP MX standardization. ¹²
- **Model parallelism** (tensor parallel and/or expert parallel for MoE, sometimes pipeline parallel) to distribute weights and activations; vendor narratives increasingly emphasize topology-aware collectives and embedding those collectives into execution graphs to reduce host overhead and tail latency. ¹³
- **Disaggregated serving** (prefill on one resource pool, decode on another) when traffic patterns justify it. This is explicit in the Cerebras-AWS Trainium/WSE collaboration description and in NVIDIA Dynamo/TensorRT-LLM design docs. ¹⁴

Vendor architectures and systems

The descriptions below focus on the items you requested: compute array style, memory hierarchy, interconnect/topology, sparsity/quantization, batching/sequence length handling, model parallelism, compiler/runtime, and practical tradeoffs.

Cerebras wafer-scale + memory-disaggregated systems

Compute array type and execution model. Cerebras' published architecture deep dive describes many small cores connected by a wafer-scale fabric, with explicit emphasis on dataflow scheduling and fine-grained mechanisms (including native support for unstructured sparsity in the WSE-2 era). ¹⁵ While WSE-3 details differ by generation, the WSE-3 positioning continues the same themes: "single device simplicity" at extreme scale. ¹⁶

On-chip memory hierarchy. WSE-3-based materials prominently claim **44 GB on-chip memory (SRAM)** and extremely high internal bandwidth (e.g., **21 PB/s memory bandwidth** and **214 PB/s core-to-core** in one official cluster datasheet). ¹⁷ The earlier architecture paper explains the rationale: distribute SRAM close to compute, avoiding cache/DRAM-centric bottlenecks, with explicit per-core SRAM and fabric-mediated sharing. ¹⁵

Off-chip memory and "MemoryX." Cerebras also describes a disaggregated external memory subsystem ("MemoryX") for model weight storage at large scale, paired with an off-chip scale-out fabric ("SwarmX") that broadcasts weights and reduces gradients in training contexts. ¹⁸ Even where the public PDF emphasizes training, Cerebras' more recent system messaging uses the same disaggregation idea for inference (splitting phases or isolating memory-heavy work). ¹⁹

Interconnect/topology. In the earlier deep dive, SwarmX is described as a tree topology interconnect designed to enable "data-parallel-only" scaling (for training) while reducing software complexity. ²⁰ For inference, Cerebras instead emphasizes eliminating weight fetch from off-chip memory during decode by keeping weights on-chip, using wafer-scale internal fabric bandwidth rather than multi-GPU collectives as the primary "scale-up" mechanism. ²¹

Quantization, sparsity, and batching/sequence length. Cerebras publicly emphasizes **full-accuracy runs** in several inference announcements (e.g., "16-bit weights" for Llama 3.1 405B in a record claim), but also explicitly uses **speculative decoding** in production-facing updates (which shifts the throughput/latency frontier by changing decode dynamics). ²²

Reported inference performance. Cerebras publishes multiple model-facing metrics that are unusually concrete for a hardware vendor:

- Llama 3.1 70B: **2,100 tokens/s** claimed after a software update. ²³
- Llama 3.1 405B: **969 output tokens/s**, **~240 ms TTFT**, and **128K context** support in a reported record run, with third-party benchmark references. ²⁴
- Ongoing positioning for "instant speed" inference is explicitly tied to the "weights on-chip SRAM" thesis and bandwidth math that argues GPU servers cannot reach the implied bandwidth targets for "instantaneous" decode at large model sizes. ²⁵

Power/area/cost tradeoffs. Cerebras positions wafer-scale as enabling extreme on-chip memory bandwidth and simplified programming (fewer distributed training/inference pathologies), at the cost of a nonstandard form factor and unusually high integration complexity (yield, cooling, power delivery). The company also publicly discusses wafer-scale yield strategies as a core enabler of manufacturability at very large die area. ²⁶

NVIDIA GPUs + NVLink rack-scale systems + inference software stack

Compute array type. NVIDIA's approach remains GPU-centric: many SMs with specialized Tensor Cores, backed by high-bandwidth memory and a mature software stack. The key shift for LLM inference has been adding *low-precision acceleration* and *system-level scale-up* that makes a rack behave like a “single massive GPU.” ²⁷

Memory hierarchy and quantization. NVIDIA's public Blackwell positioning emphasizes a second-generation Transformer Engine that enables **FP4 inference** (and microscaling formats), explicitly stating that it increases the effective model sizes supported by memory while maintaining accuracy. ²⁷ In the Blackwell platform press materials, NVIDIA also highlights a **10 TB/s** chip-to-chip link between two GPU dies and positions the architecture for models “scaling up to 10 trillion parameters.” ²⁸

Interconnect/topology. NVIDIA's Blackwell materials highlight:

- **~130 TB/s** of GPU communications inside a 72-GPU NVLink domain (NVL72). ²⁷
- **Scaling NVLink** to large GPU counts (e.g., a stated ability to scale to hundreds of GPUs for large models, with per-GPU NVLink bandwidth figures in press materials). ²⁹

Compiler/runtime: TensorRT-LLM + Dynamo. NVIDIA's differentiation for inference at scale is increasingly in the *software orchestration layer*:

- TensorRT-LLM documentation describes **paged KV cache**, request scheduling (“in-flight batching/continuous batching”), long-sequence support, quantization options, parallelism schemes, and disaggregated serving modes. ³⁰
- NVIDIA Dynamo is described as an open-source distributed inference framework that coordinates backends (vLLM/SGLang/TensorRT-LLM), enabling **intelligent routing, disaggregated prefill/decode**, and **multi-tier KV caching** for data center-scale inference. ³¹

Measured/reported performance touchpoints. NVIDIA's own public pages emphasize relative speedups such as “30× real-time inference” comparisons (with stated benchmark conditions) for GB200 NVL72 vs H100 infrastructure and strong energy-efficiency claims (noting projected performance disclaimers). ³² A useful “real deployment” datapoint comes from Baseten ³³, which reports **~650 tokens/s on GPT-OSS 120B on B200 GPUs** with a stack including TensorRT-LLM, NVIDIA Dynamo, TP8, and EAGLE-3 speculative decoding, citing independent benchmarking. ³⁴

Practical tradeoff profile. NVIDIA's strength is breadth: any model family, continual kernel/runtime optimization, and ecosystem tooling. The cost is that decode often remains fundamentally constrained by the memory hierarchy and by multi-device communication overhead as models and contexts scale—hence the emphasis on disaggregation, KV-cache-aware routing, and topology-aware collectives in software. ³⁵

Groq LPU: deterministic, compiler-scheduled token pipelines

Compute array type. Groq's LPU is described as a "programmable assembly line" with data "conveyor belts" that stream instructions and data between SIMD function units; the emphasis is on eliminating dynamic scheduling contention. ³⁶

On-chip memory hierarchy. Groq claims **hundreds of MB of SRAM** as *primary weight storage* (not cache), with a public spec sheet citing **230 MB SRAM per chip** and **up to 80 TB/s on-die memory bandwidth**. ³⁷

Interconnect/topology. Groq describes direct chip-to-chip connectivity via a plesiosynchronous protocol and emphasizes that the compiler predicts data arrival timing to coordinate compute + network schedules. The GroqChip spec sheet cites **16 integrated chip-to-chip interconnects** (RealScale). ³⁸

Numerics and quantization. Groq's larger-model blog emphasizes "TruePoint" numerics (including high-precision accumulation) and compiler-controlled mixed precision choices (e.g., selectively using higher precision where errors propagate). ³⁹

Model parallelism strategy. Groq explicitly frames **tensor parallelism** as the key latency reducer for real-time inference, enabled by on-chip SRAM weight storage and deterministic timing to avoid tail-latency amplification across collectives. It also claims to layer pipeline parallelism on top of tensor parallelism via static scheduling. ⁴⁰

Public performance metrics. Groq's publicly accessible *chip* specs include **up to 750 TOPS INT8 / 188 TFLOPs FP16** at a stated frequency and power figures (TDP and max power) for a 14nm part. ⁴¹ Model-specific tokens/s for frontier-scale LLMs are less consistently published in primary materials than the Cerebras reports; however, Groq explicitly cites running "Kimi K2" as a "1-trillion-parameter model" on GroqCloud in a technical blog, which is directly relevant to your 1T-class target. ³⁹

d-Matrix Corsair: digital in-memory compute chiplets for decode

Compute array type. d-Matrix positions Corsair around "Digital In-Memory Compute (DIMC)," using SRAM arrays and bit-serial/bit-parallel MAC structures, plus vector/SIMD components for flexibility (as described in Hot Chips materials). ⁴²

Memory hierarchy. Public materials include:

- **2 GB SRAM** and **150 TB/s** (bandwidth figure given "per PCIe" in the Hot Chips slide deck). ⁴³
- **256 GB LPDDR5** with **~400 GB/s**, and a chiplet-based organization (8 chiplets), with a stated **600 W TDP**. ⁴⁴

Interconnect/topology. d-Matrix emphasizes low-latency collectives across chiplets and packages, describing hierarchical all-to-all patterns and scale-out via a "Transparent NIC" approach that transports PCIe semantics over Ethernet, aiming to reduce latency relative to conventional RDMA paths. ¹¹

Quantization/compression. d-Matrix emphasizes block floating point numerics (aligned with OCP MX framing) and advertises substantial weight compression (e.g., "supports 5× weight compression" in the Hot Chips deck). ⁴⁵

Serving/runtime. d-Matrix's software stack ("Aviator") is described as embedding topology-aware collectives into execution graphs, minimizing host involvement, and supporting dynamic shapes and on-device autoregression (important for decode). ⁴⁶

Reported performance metrics. d-Matrix publishes (1) comparative claims vs H100 and (2) explicit *token/s projection tables* under specified conditions:

- The whitepaper includes comparative figures and describes order-of-magnitude time-per-token improvements in selected scenarios, and
- It provides a table projecting **~60,000 tokens/s** for Llama3 8B and **~30,000 tokens/s** for Llama3 70B at specific input/output lengths and batch sizes, with **~1-2 ms/token** time-per-output-token in those projections. ⁴⁷

The same whitepaper also claims **~3× energy efficiency** and **~3× cost-performance** advantages over "GPU alternatives" and "10× higher interactivity," but these are presented as vendor claims rather than independently standardized benchmarks. ⁴⁸

Etched transformer-only ASICs: extreme specialization, limited verifiable detail

Etched's public positioning is that Sohu is a transformer-only ASIC, trading generality for utilization and throughput. Available information in broadly accessible primary/near-primary sources is dominated by claims rather than detailed architecture disclosures.

- Press coverage reports that Sohu is transformer-only, and attributes claims such as "one Sohu server replaces 160 H100 GPUs," with manufacturing/process and performance claims described as company statements. ⁴⁹
- Etched's own public announcement posts (for example, on X) frame Sohu as a transformer-specialized ASIC and promote large performance improvements, but without the type of detailed memory hierarchy, topology, or reproducible benchmark methodology that appears in the other vendors' materials. ⁵⁰

This creates a practical research gap: without detailed published microarchitecture + measured silicon results, it is difficult to rigorously compare Etched to systems that publish concrete specs, software architecture, and model-facing benchmarks.

Prioritized source links

Cerebras WSE-3 / Wafer-Scale Cluster datasheet (PDF)
<https://cdn.sanity.io/files/e4qjo92p/production/b3779b8136a3a03a7b31913cb680e33d68423e58.pdf>

Cerebras inference benchmarks (Llama 70B / 405B)
<https://www.cerebras.ai/blog/cerebras-inference-3x-faster>
<https://www.cerebras.ai/blog/llama-405b-inference>

Cerebras + AWS disaggregated inference announcement
<https://www.cerebras.ai/blog/cerebras-is-coming-to-aws>

Cerebras architecture deep dive (IEEE Micro / Hot Chips 34) (PDF)
<https://8968533.fs1.hubspotusercontent-na1.net/hubfs/8968533/IEEE%20Micro%202023-03%20Hot%20Chips%2034%20Cerebras%20Architecture%20Deep%20Dive.pdf>

NVIDIA Blackwell platform press release (PDF)
https://nvidianews.nvidia.com/_gallery/download_pdf/65f8a7843d633205563719fc/

NVIDIA Blackwell architecture page + GB200 NVL72 page
<https://www.nvidia.com/en-us/data-center/technologies/blackwell-architecture/>
<https://www.nvidia.com/en-us/data-center/gb200-nvl72/>

TensorRT-LLM docs
<https://nvidia.github.io/TensorRT-LLM/>

NVIDIA Dynamo overview + docs
<https://developer.nvidia.com/dynamo>
<https://docs.nvidia.com/dynamo/getting-started/introduction>

Groq LPU explained + LPU architecture page
<https://groq.com/blog/the-groq-lpu-explained>
<https://groq.com/lpu-architecture>

GroqChip spec sheet (PDF)
<https://cdn.sanity.io/files/cho10sk5/production/908a9db1814f76a4a5c16ac63cc064058460376b.pdf>

d-Matrix Corsair technical whitepaper (PDF) + Hot Chips deck (PDF)
<https://d-matrix.ai/pdf/d-Matrix-WhitePaper-Technical-FINAL.pdf>
<https://www.d-matrix.ai/wp-content/uploads/2025/10/hc2025.dmatrix.SudeepBhoja.v01.pdf>

Etched (press coverage + official post)
<https://techcrunch.com/2024/06/25/etched-is-building-an-ai-chip-that-only-runs-transformer-models/>
<https://x.com/Etched/status/1805625693113663834>

Performance metrics and scalability to 1T-10T inference

Cross-vendor comparison is inherently noisy because vendors report different models, precisions, batching, and context lengths. The tables below therefore separate (a) **architectural capacity/bandwidth envelopes** from (b) **model-facing tokens/s claims**.

Architecture feature comparison

Platform	Compute style	On-chip memory	Off-chip memory	Scale-up / interconnect	Parallelism + serving stack highlights
Cerebras (WSE-3 / CS-3 class)	Wafer-scale manycore + on-wafer fabric	44 GB SRAM; PB/s-scale internal BW claimed ¹⁷	“MemoryX” external memory SKUs cited; cluster/network specs include 100GbE ports and large off-chip capacity ⁵¹	Wafer internal fabric dominates; SwarmX discussed as tree fabric in earlier architecture for scaling simplicity ²⁰	Emphasis on removing GPU-like decode bandwidth wall; speculative decoding appears in inference releases ⁵²
NVIDIA (GB200 NVL72 / Blackwell-era)	GPU SMs + Tensor Cores	Cache/SRAM not the primary narrative; focus is HBM + system fabrics	Rack-scale memory pool framing (“fast memory” / NVLink-C2C / Grace) and low-precision to increase effective capacity ⁵³	NVLink Switch System: ~130 TB/s in a 72-GPU domain; scale to large GPU counts for trillion-parameter models ²⁷	TensorRT-LLM (paged & quantized KV, IFB/continuous batching, disaggregation) + Dynamo orchestration (KV-aware routing, multi-tier KV caching) ⁵⁴
Groq (GroqChip / LPU)	Deterministic “assembly line” streaming SIMD	230 MB SRAM, ~80 TB/s on-die BW (spec sheet) ⁴¹	No HBM focus; SRAM is positioned as primary weight store	Direct chip-to-chip protocol; RealScale connectors (16 on chip) ³⁸	Compiler statically schedules compute + comm; emphasizes latency-optimized tensor parallelism; positions itself for 1T-class models (Kimi K2 example) ⁴⁰

Platform	Compute style	On-chip memory	Off-chip memory	Scale-up / interconnect	Parallelism + serving stack highlights
d-Matrix (Corsair)	DIMC + vector/SIMD; chiplet platform	2 GB SRAM; large internal BW claims ⁴³	256 GB LPDDR5; ~400 GB/s; weight compression emphasized ⁵⁵	Hierarchical all-to-all, scale-up via PCIe, scale-out via “Transparent NIC” concept ¹¹	“Aviator” stack: in-graph collectives, dynamic shapes, on-device autoregression; explicit focus on token-generation bottlenecks ¹¹
Etched (Sohu)	Transformer-only ASIC (claimed)	Not publicly detailed in primary docs; mostly performance claims	Not publicly detailed in primary docs; press describes a transformer-only inference pipeline	Not publicly detailed; press claims focus on server-level replacement ratios	Claims large throughput/efficiency gains; limited published microarchitecture detail → difficult to verify rigorously ⁶

Reported model-facing metrics (tokens/s, TTFT, etc.)

Platform	Model / scenario	Precision / settings (as reported)	Reported metric(s)	Notes on comparability
Cerebras	Llama 3.1 70B	Update introduces speculative decoding; output speed cited as “average” ⁵⁶	2,100 tokens/s ⁵⁷	Vendor-reported; context/batch conditions matter; still unusually concrete compared to typical hardware marketing.
Cerebras	Llama 3.1 405B	“16-bit weights,” 128K context cited in record run ²⁴	969 tokens/s, ~240 ms TTFT, 128K context ²⁴	Claims reference third-party benchmarking; still not directly comparable to other vendors’ setups.
NVIDIA GPUs (via Baseten)	GPT-OSS 120B on B200	TP8 EP1 across 8×B200; TensorRT-LLM + Dynamo; EAGLE-3 speculative decoding ³⁴	~650 tokens/s reported uplift (from ~400) ⁵⁸	Useful as a “production engineering” datapoint for Blackwell-class GPUs.

Platform	Model / scenario	Precision / settings (as reported)	Reported metric(s)	Notes on comparability
NVIDIA (system claim)	GB200 NVL72 (trillion-parameter inference)	NVIDIA marketing benchmark conditions provided; projected performance subject to change ³²	"30× real-time LLM inference vs H100" (headline) ³²	A relative claim; interpret as directional absent standardized tokens/s workloads.
d-Matrix	Llama3 8B	MXINT8; 1024/1024; batch 48 (table) ⁵⁹	~60,000 tokens/s, ~1 ms/token ⁵⁹	Explicitly labeled "projections"; depends strongly on batching and target latency constraints.
d-Matrix	Llama3 70B (rack-scale)	MXINT8; 1024 input / 3072 output; batch 64 (table) ⁵⁹	~30,000 tokens/s, ~2 ms/token ⁵⁹	Projection table; note this is <i>aggregate</i> in their described rack scenario.
Groq	Kimi K2 (1T parameters)	Not fully specified in the blog excerpt; positioned as "preview" on GroqCloud ³⁹	No single canonical tokens/s number in the cited primary sources	Chip-level TOPS, SRAM, and deterministic execution are documented, but model-specific public throughput varies by workload and publication. ⁶⁰

Implications for scaling to 1T–10T models

- **Single-device inference is generally infeasible** for dense 1T weights unless the device has multi-terabyte memory (or the model is heavily quantized and/or sparsified). GPU NVLink domains provide large aggregate memory, while wafer-scale approaches attempt to keep as much as possible close to compute for decode. ⁶¹
- **1T–10T pushes you into explicit distributed inference**, and the "hard part" becomes orchestration: avoiding tail latency while doing tensor/expert parallel collectives, managing KV cache, and routing traffic based on cache locality. NVIDIA Dynamo and the Baseten engineering writeup both emphasize these themes (KV-aware routing, disaggregation, and linking parallelism settings to SLA targets). ⁶²
- **Disaggregation is not optional at the extreme**: separating prefill and decode enables hardware specialization and capacity matching, as seen in the Amazon Web Services ⁶³ + Cerebras Trainium/ WSE collaboration narrative and in NVIDIA's Dynamo/TensorRT-LLM design approach. ¹⁴

Pros, cons, and gaps for very large LLM inference

Cerebras

Advantages: The “weights on-chip” thesis targets the decode bottleneck directly; published tokens/s and TTFT numbers demonstrate user-facing speed at substantial model sizes; internal fabric bandwidth and SRAM capacity claims are exceptional. ⁶⁴

Limitations/gaps: For 1T–10T, even wafer-scale SRAM will not hold full dense weights; scaling beyond a few devices still requires system-level orchestration, and vendor-published inference benchmarks are not a substitute for standardized, reproducible cluster-level measurements at 1T+ with long-context concurrency. ⁶⁵

NVIDIA

Advantages: Best-in-class ecosystem maturity and flexibility; rapid incorporation of new inference techniques (disaggregation, paged KV cache, speculative decoding, KV reuse/routing) into open documentation and open-source projects; proven ability to scale networks and provide rack-scale NVLink domains for trillion-parameter inference narratives. ⁶⁶

Limitations/gaps: Even with NVLink, large distributed inference increasingly becomes a cluster scheduling and memory-management problem; achieving “interactive” latency at long context and large batch requires complex tuning (parallelism configuration, KV cache policies, routing, and often disaggregation). NVIDIA itself increasingly treats these as first-class software problems, which is an implicit admission that hardware alone is insufficient. ⁶⁷

Groq

Advantages: Deterministic execution and static scheduling can reduce tail latency; on-die SRAM as primary store plus direct interconnect is a coherent design for low-latency decode at moderate per-chip memory footprints; the company explicitly emphasizes tensor-parallel latency reduction and showcases a 1T-class model example. ⁶⁸

Limitations/gaps: The publicly documented SRAM capacity per chip is far below what dense 1T weights require, so large-model inference implies heavy sharding and/or quantization plus careful pipeline design; primary public materials are stronger on architectural philosophy and chip specs than on repeatable, apples-to-apples large-model throughput/latency benchmarks across full context and realistic traffic mixes. ⁶⁹

d-Matrix

Advantages: DIMC + chiplets directly targets memory bandwidth and energy efficiency for decode; openly published system specs (SRAM, LPDDR, chiplets) and token/s projection tables; explicit attention to hybrid parallelism and embedding collectives into graphs, which is critical at scale. ⁷⁰

Limitations/gaps: Many performance numbers are described as preliminary or projections; LPDDR-based designs have a different latency/bandwidth trade space than HBM-based systems and may face scaling

challenges for the largest dense models unless compression, MoE sparsity, or multi-rack orchestration is exceptionally effective. ⁷¹

Etched

Advantages (in principle): Extreme transformer specialization could increase utilization and reduce overhead for fixed workloads compared to general GPUs. ⁶

Limitations/gaps: Compared with other vendors, there is a major shortage of publicly available primary technical material (memory hierarchy, interconnect, compiler/runtime details, measured silicon performance under reproducible conditions). This makes rigorous assessment for 1T–10T inference impossible from public data alone. ⁶

Proposed hybrid architecture

Design goal

A practical 1T–10T inference system must (a) disaggregate prefill and decode, (b) treat KV cache as a first-class “memory tiering + routing” problem, and (c) sustain large-scale tensor/expert parallelism without tail-latency explosions. Current vendor approaches expose strong components of this solution—wafer-scale SRAM for decode, NVLink rack-scale domains + mature runtime software, deterministic token pipelines, and DIMC/weight compression—but none fully closes the loop alone at the 10T extreme. ⁷²

Proposed architecture: Hybrid Disaggregated Inference Fabric

This is a *system* + *ASIC* concept rather than a single chip:

- **Prefill ASIC (“P-chip”)**: compute-dense, HBM-backed, optimized for large GEMMs and high throughput (similar to training-class acceleration), supports FP8/FP4-like low precision and high batch.
- **Decode ASIC (“D-chip”)**: SRAM-heavy near-compute, optimized for streaming weight/KV access and deterministic token-step scheduling for low TPOT and low jitter.
- **KV Cache Fabric (“K-fabric”)**: a memory-tier + routing layer that keeps “hot” KV on-package (HBM/SRAM), and spills “warm/cold” KV to a pooled memory tier with fast DMA and compression; request routing is KV-prefix aware.
- **Topology-aware collectives**: hardware support for AllReduce / AlltoAll patterns common in tensor/expert parallelism, with execution-graph embedding (minimize host involvement), borrowing from both d-Matrix and modern GPU runtime patterns. ⁷³

Block diagram

flowchart TB

```
User[Client / API Gateway] --> Router[KV-aware Request Router]
Router -->|prefill| PrefillPool[Prefill Pool: P-chip nodes]
Router -->|decode| DecodePool[Decode Pool: D-chip nodes]
```

```

PrefillPool -->|KV cache transfer| KVXfer[KV Cache Transfer Layer]
KVXfer --> DecodePool

subgraph PrefillPool[Prefill Pool]
  P1[P-chip] --> PHBM[HBM + SRAM cache]
  P2[P-chip] --> PHBM2[HBM + SRAM cache]
end

subgraph DecodePool[Decode Pool]
  D1[D-chip] --> DSRAM[Large on-chip SRAM + stream engine]
  D2[D-chip] --> DSRAM2[Large on-chip SRAM + stream engine]
end

subgraph KVFabric[KV Cache Fabric]
  Hot[Hot KV Tier: HBM/SRAM] --> Warm[Warm KV Tier: pooled DRAM/CXL-like]
  Warm --> Cold[Cold KV Tier: object store / SSD]
end

PrefillPool -->|evict/promote| KVFabric
DecodePool -->|reuse/promote| KVFabric
Router -->|prefix match signals| KVFabric

DecodePool --> Collectives[Topology-aware Collectives\n(AllReduce / AlltoAll)]
PrefillPool --> Collectives

```

Target specs (order-of-magnitude goals)

These are *engineering targets* derived from constraints highlighted in vendor materials (memory-bound decode, long-context KV scaling, and disaggregation), not vendor-validated measurements. ⁷⁴

- **D-chip:**
 - On-chip SRAM: 64–256 GB-equivalent via multi-die SRAM chiplets (aggressive, but aligned with the “weights close to compute” direction of multiple vendors).
 - Deterministic token-step engine (static schedule for the decode critical path).
 - Hardware KV cache compression/decompression and paged KV management (conceptually similar to paged KV designs in TensorRT-LLM, but pushed into firmware/hardware). ⁷⁵
- **P-chip:**
 - HBM-rich, FP8/FP4 optimized prefill throughput; designed to saturate large batch prompt traffic. ²⁷
- **Fabric goals:**
 - Intra-rack scale-up bandwidth on the order of NVL72-class fabrics (100+ TB/s aggregate within a rack-scale domain) to keep expert/tensor parallel collectives from dominating latency. ²⁷
 - KV transfer path optimized for disaggregated serving (explicit KV transmission semantics). ⁷⁶
- **Performance targets** (system-level):
 - 1T model: sub-second “time-to-result” for typical interactive prompts (achieved by disaggregation + speculative decoding + deterministic decode pipeline).
 - 10T model (likely MoE): interactive TTFT with bounded TPOT by isolating expert AlltoAll on high-bandwidth scale-up fabrics and routing requests to maximize KV reuse.

Why this improves on observed weaknesses

- It treats **prefill vs decode** as *different hardware problems*, consistent with both the Cerebras–AWS narrative and NVIDIA Dynamo/TensorRT-LLM design docs. ¹⁴
- It treats **KV cache** as a *tiered memory + routing* problem, consistent with NVIDIA's KV cache reuse / KV-aware routing directions and with the cluster-level emphasis in Baseten's production writeup. ⁷⁷
- It acknowledges that **communication patterns** (AllReduce/AlltoAll) are now first-order for MoE and large sharded models, matching NVIDIA's and d-Matrix's emphasis on topology-aware collectives and high-bandwidth scale-up. ⁷⁸

Implementation plan, orchestration emulation, and phased roadmap

Practical engineering plan using coding tools + hardware design flows

The fastest path to a credible prototype is to build a *full-stack emulator first*, then iteratively “harden” the most performance-critical pieces into RTL and hardware prototypes.

Software and systems foundation (start immediately).

- Build a serving stack that supports: disaggregated prefill/decode, KV-cache-aware routing, paged KV cache allocations, and topology-aware parallelism settings. NVIDIA's public docs clearly define these primitives (Dynamo design docs + TensorRT-LLM features). ⁷⁹
- Implement a “router” service that makes placement decisions using prefix/KV locality signals, similar in concept to what production providers describe when optimizing TTFT and throughput. ⁸⁰

Use coding agents to accelerate the build/test loop.

You can use OpenAI ⁸¹ Codex as a multi-agent coding workflow tool (parallel tasks, sandboxed environments, codebase understanding) to generate boilerplate services, benchmarking harnesses, and CI pipelines, while keeping human review on performance-critical and security-sensitive components. ⁸²

Hardware design flow (ASIC + chiplets).

- **Architecture modeling:** cycle-level or event-driven simulator for token pipelines, KV movement, and collective operations; validate against known behaviors reported by vendors (e.g., token-generation memory-bound regime and SLO-driven small batches). ⁸³
- **RTL/HLS split:** implement the decode-critical datapath (KV read + attention + matmul + quant/format conversion) as RTL, while using HLS for less critical DMA/control blocks early.
- **Verification:** heavy property checking and constrained-random verification for DMA, coherence-free paging, and collective primitives. (This is where most advanced-node schedule risk typically concentrates, consistent with industry cost discussions.) ⁸⁴
- **FPGA prototyping:** emulate the deterministic token-step engine + KV paging + router hooks on FPGA boards while using host DRAM/HBM proxies for memory-tier behavior.

Emulating “startup-style” approaches: sharding, orchestration, autotuning

Public information about “Recursive Intelligence” as an inference orchestration startup is not strongly evidenced in broadly available primary technical sources; the most visible “Recursive/Ricursive” entity in recent reporting focuses on AI-assisted chip design automation rather than inference serving. ⁸⁵ Nevertheless, the *approach you asked for*—model sharding + orchestration + autotuning—can be implemented concretely using patterns already documented in modern inference stacks:

- **Model sharding primitives:** implement tensor parallel (TP), expert parallel (EP) for MoE, and optional pipeline parallel (PP) in the serving runtime; expose them as tunables. ⁸⁶
- **Orchestration layer:** an operator/controller that (a) decides aggregated vs disaggregated deployment, (b) selects `#prefill_workers` and `#decode_workers`, and (c) sets TP/EP/PP parameters to meet an SLA. NVIDIA Dynamo explicitly documents an “AIConfigurator” concept that searches these design points automatically. ⁸⁷
- **Autotuning loop:** run closed-loop experiments on real traffic traces, optimizing an objective like `(throughput subject to  $TTFT \leq X$  and  $TPOT \leq Y$ )`. Baseten’s description of iterating speculative decoding variants and routing strategies is a concrete template for such a loop. ⁵⁸

Risks and cost (order-of-magnitude)

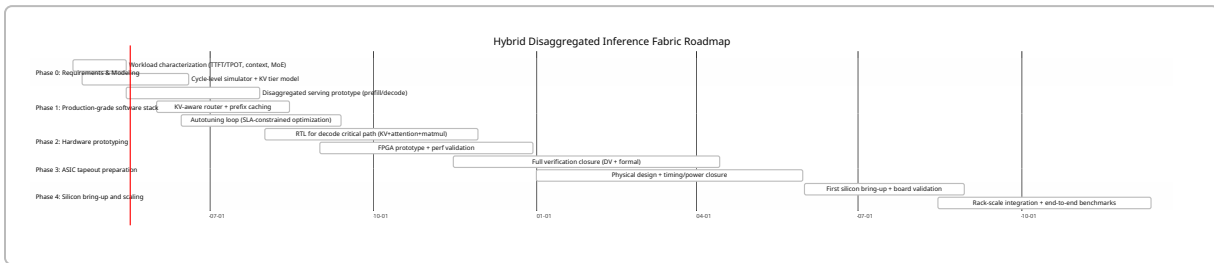
Major technical risks

- **Memory system risk:** the KV cache + long-context regime can dominate both capacity and bandwidth; mis-sizing memory tiers or paging policies can wipe out the theoretical advantage of a custom decode ASIC. ⁸⁸
- **Scale-out tail latency:** as you increase TP/EP/PP degrees for 1T–10T models, collective operations can dominate token time unless the topology and runtime are co-designed. NVIDIA and d-Matrix both highlight topology-aware collectives and high-bandwidth scale-up as essential. ⁷⁸
- **Algorithm drift risk:** transformer-only specialization (Etched-style) is strategically risky if attention mechanisms or model families shift; this is a core tradeoff of extreme specialization highlighted by the nature of the product positioning in available sources. ⁶

Cost ranges (very approximate)

- Advanced-node custom silicon is routinely discussed in industry analysis as having very large non-recurring engineering (NRE) costs (hundreds of millions at leading nodes when accounting for design, verification, and masks), with substantial uncertainty depending on node, die size, and IP reuse. ⁸⁹
- Advanced packaging is a first-order constraint and cost driver for AI accelerators; TSMC ⁹⁰ positions CoWoS as a key technology for integrating logic chiplets with HBM for AI/HPC workloads, and trade press notes scaling interposer sizes and HBM stack counts over time. ⁹¹
- Reticle limits constrain monolithic die size and push designs toward multi-die packaging or wafer-scale strategies; industry sources commonly cite exposure-field constraints in the tens of mm range, which maps to sub-1,000 mm² reticle-limited dies. ⁹²

Phased roadmap and milestones



This roadmap is intentionally staged so that the *core product value* (high-speed, SLA-constrained disaggregated inference with KV-aware routing and autotuning) is demonstrated in software first—mirroring the direction emphasized by modern inference frameworks—before committing to the full cost and risk of a leading-edge tapeout. ⁹³

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